

YOLO模型训练

本次案例使用模型为YOLOV5-Lite

github地址为:[ppogg/YOLOv5-Lite: 🍌🍌🍌 YOLOv5-Lite: Evolved from yolov5 and the size of model is only 900+kb \(int8\) and 1.7M \(fp16\). Reach 15 FPS on the Raspberry Pi 4B~ \(github.com\)](https://github.com/ppogg/YOLOv5-Lite).

将我们的模型解压到任意地址,解压后复制文件地址

如果你的电脑不是英伟达显卡:

我们打开Anaconda Prompt输入:

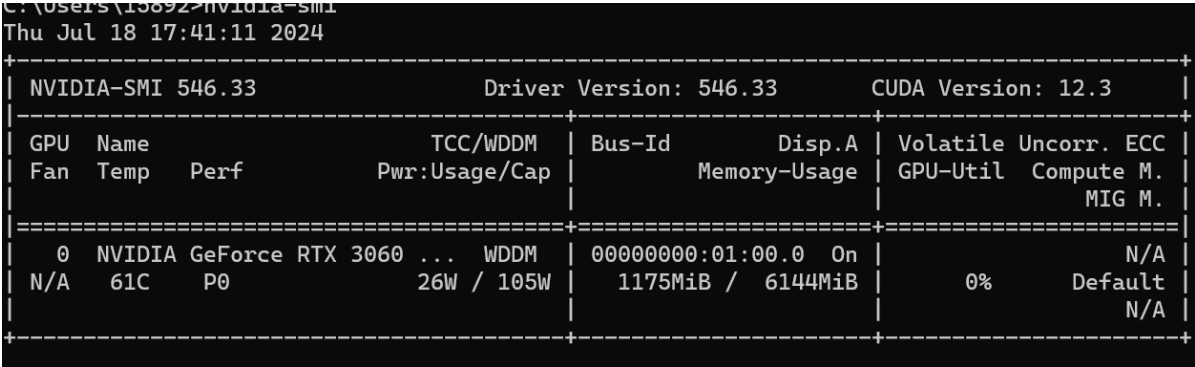
```
cd 你的模型文件地址
conda create -n yolov5-lite python=3.10
pip install -r requirements.txt
pip install numpy<2
```

注意,如果发现 pip install 安装较慢请输入以下指令进行换源:

```
python -m pip install --upgrade pip
pip config set global.index-url https://pypi.tuna.tsinghua.edu.cn/simple
```

如果你的电脑是英伟达显卡,且你想加速模型训练过程又不怕麻烦的人:

打开CMD窗口,输入 `nvidia-smi`



可以看到我的CUDA版本为 12.3

之后我们打开英伟达 CUDA网址: <https://developer.nvidia.com/cuda-toolkit-archive>

选择一个比你显卡驱动低的CUDA版本(这里我演示11.8版本):

CUDA Toolkit Archive

Previous releases of the CUDA Toolkit, GPU Computing SDK, documentation and developer drivers can be found using the links below. Please select the release you want from the list below, and be sure to check www.nvidia.com/drivers for more recent production drivers appropriate for your hardware configuration.

[Download Latest CUDA Toolkit](#)

[Learn More about CUDA Toolkit](#)

Latest Release

[CUDA Toolkit 12.3.1](#) (November 2023), [Versioned Online Documentation](#)

Archived Releases

[CUDA Toolkit 12.3.0](#) (October 2023), [Versioned Online Documentation](#)
[CUDA Toolkit 12.2.2](#) (August 2023), [Versioned Online Documentation](#)
[CUDA Toolkit 12.2.1](#) (July 2023), [Versioned Online Documentation](#)
[CUDA Toolkit 12.2.0](#) (June 2023), [Versioned Online Documentation](#)
[CUDA Toolkit 12.1.1](#) (April 2023), [Versioned Online Documentation](#)
[CUDA Toolkit 12.1.0](#) (February 2023), [Versioned Online Documentation](#)
[CUDA Toolkit 12.0.1](#) (January 2023), [Versioned Online Documentation](#)
[CUDA Toolkit 12.0.0](#) (December 2022), [Versioned Online Documentation](#)
[CUDA Toolkit 11.8.0](#) (October 2022), [Versioned Online Documentation](#)
[CUDA Toolkit 11.7.1](#) (August 2022), [Versioned Online Documentation](#)
[CUDA Toolkit 11.7.0](#) (May 2022), [Versioned Online Documentation](#)
[CUDA Toolkit 11.6.2](#) (March 2022), [Versioned Online Documentation](#)
[CUDA Toolkit 11.6.1](#) (February 2022), [Versioned Online Documentation](#)
[CUDA Toolkit 11.6.0](#) (January 2022), [Versioned Online Documentation](#)
[CUDA Toolkit 11.5.2](#) (February 2022), [Versioned Online Documentation](#)
[CUDA Toolkit 11.5.1](#) (November 2021), [Versioned Online Documentation](#)

CUDA Toolkit 11.8 Downloads

Select Target Platform

Click on the green buttons that describe your target platform. Only supported platforms will be shown. By downloading and using the software, you agree to fully comply with the terms and conditions of the [CUDA EULA](#).

Operating System

Linux

Windows

Resources

- [CUDA Documentation/Release Notes](#)
- [MacOS Tools](#)
- [Training](#)
- [Sample Code](#)
- [Forums](#)
- [Archive of Previous CUDA Releases](#)
- [FAQ](#)
- [Open Source Packages](#)
- [Submit a Bug](#)
- [Tarball and Zip Archive Deliverables](#)

CUDA Toolkit 11.8 Downloads

Select Target Platform

Click on the green buttons that describe your target platform. Only supported platforms will be shown. By downloading and using the software, you agree to fully comply with the terms and conditions of the [CUDA EULA](#).

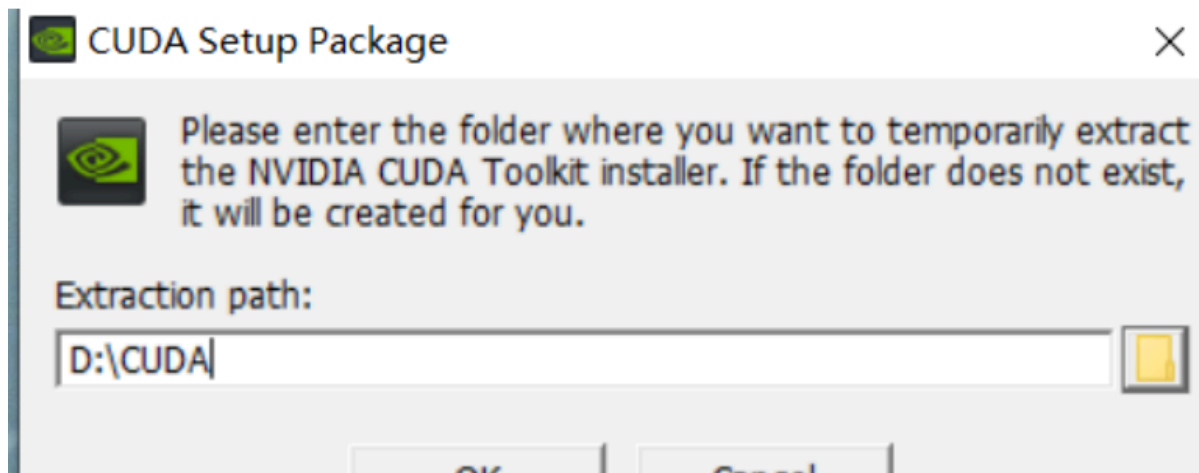
Operating System	Linux	Windows
Architecture	x86_64	
Version	10	11
Installer Type	exe (local)	exe (network)

Download Installer for Windows 11 x86_64

The base installer is available for download below.

> Base Installer	Download (3.0 GB)
------------------	-------------------

下载完后,双击安装:



选择同意,然后选择精简:



查看是否安装成功

在命令窗口中输入 `nvcc -v` 进行检查

```
nvcc -v
```

可以看到我们安装成功!

```
管理员: C:\WINDOWS\system...
Microsoft Windows [版本 10.0.22631.3085]
(c) Microsoft Corporation. 保留所有权利。

C:\Users\15892>nvcc -v
nvcc: NVIDIA (R) Cuda compiler driver
Copyright (c) 2005-2022 NVIDIA Corporation
Built on Tue_May_3_19:00:59_Pacific_Daylight_Time_2022
Cuda compilation tools, release 11.7, V11.7.64
Build cuda_11.7.r11.7/compiler.31294372_0

C:\Users\15892>
```

安装完CUDA之后我们打开Anaconda Prompt输入:

```
cd 你的模型文件地址
conda create -n yolov5-lite python=3.10
```

注意此时我们需要将requirements.txt里面的有关pytoch的内容全部删掉,避免二次下载

```
1 # pip install -r requirements.txt
2 requests
3 # base -----
4 matplotlib>=3.2.2
5 numpy>=1.18.5
6 opencv-python>=4.1.2
7 Pillow
8 PyYAML>=5.3.1
9 scipy>=1.4.1
10 tqdm>=4.41.0
11
12 # logging -----
13 tensorboard>=2.4.1
14 # wandb
15
16 # plotting -----
17 seaborn>=0.11.0
18 pandas
19 torch==2.3.0
20 torchaudio==2.3.0
21 torchvision==0.18.0
22 # export -----
23 # coremltools>=4.1
24 # onnx>=1.9.1
25 # scikit-learn==0.19.2 # for coreml quantization
26
27 # extras -----
28 thop # FLOPS computation
29 pycocotools>=2.0 # COCO mAP
30
```

注意,如果发现 pip install 安装较慢请输入以下指令进行换源:

```
python -m pip install --upgrade pip
pip config set global.index-url https://pypi.tuna.tsinghua.edu.cn/simple
pip install -r requirements.txt
pip install numpy<2
```

接着,我们去pytorch官网那个里面下载GPU版的pytorch 官网链接:[PyTorch](https://pytorch.org/)

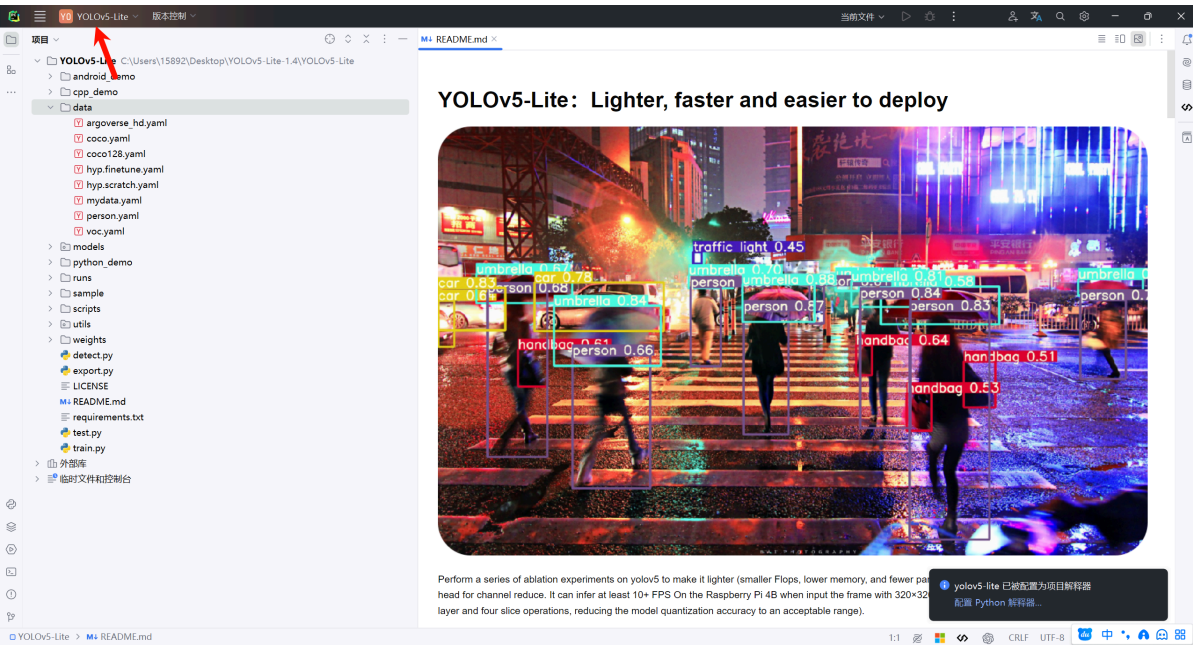
NOTE: Latest PyTorch requires Python 3.8 or later.

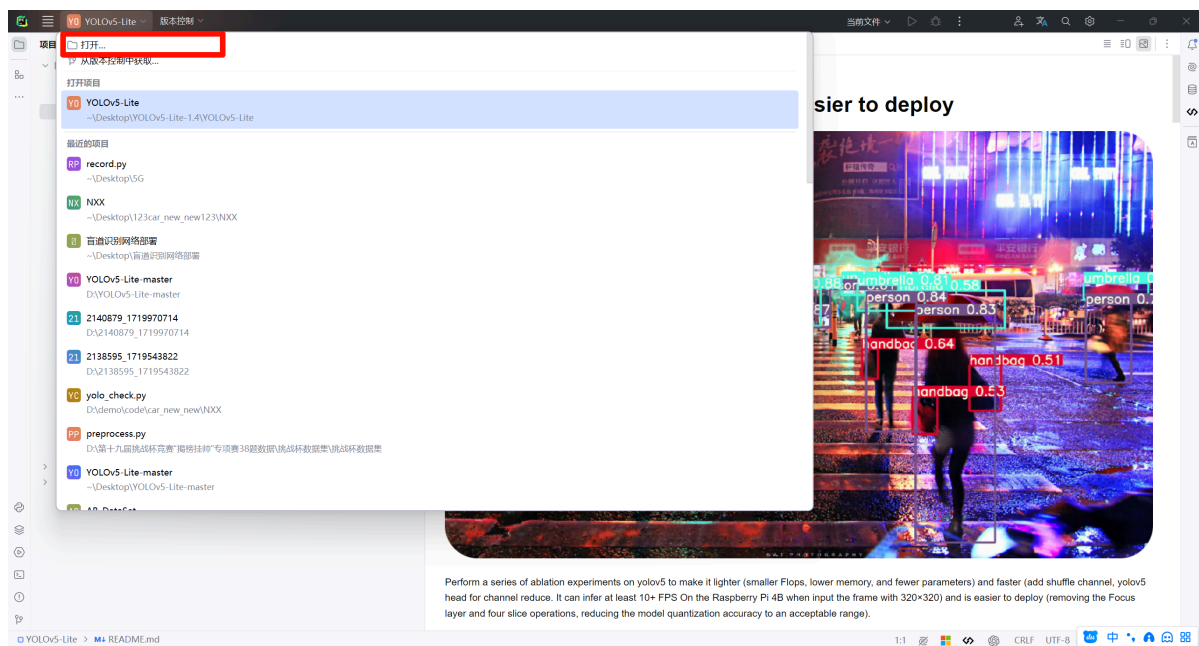
PyTorch Build	Stable (2.3.1)		Preview (Nightly)	
Your OS	Linux	Mac	Windows	
Package	Conda	Pip	LibTorch	Source
Language	Python		C++ / Java	
Compute Platform	CUDA 11.8	CUDA 12.1	CUDA 12.4	ROCm 6.0
Run this Command:	pip3 install torch torchvision torchaudio --index-url https://download.pytorch.org/whl/cu118			

选择如下选项,复制这段文字运行,等待安装好后,环境配置完成.

模型训练

安装完成之后,我们打开Pycharm点击右上角,打开我们的模型文件:





打开之后,切换Python环境:



Python 解释器

- ☐ KGGPT
 - ☐ chatglm3 (2)
 - ☐ chatglm3 (3)
 - ☐ paddle
 - ☐ yolov5-t
 - ☐ yolov8
 - ☐ yolov5-lite
 - ☐ yolo-fast
-
- ☐ test1
 - ☐ C:\Users\15892\anaconda3
 - ☐ C:\Users\15892\anaconda3 (2)
 - ☐ C:\Users\15892\anaconda3 (3)
 - ☐ IDV
 - ☐ langchain-chatglm2
 - ☐ langchain-chatglm2 (2)

🏠 添加本地解释器...

SSH...

Vagrant...

WSL...

Docker...

Docker Compose...

添加新的解释器 >

解释器设置...

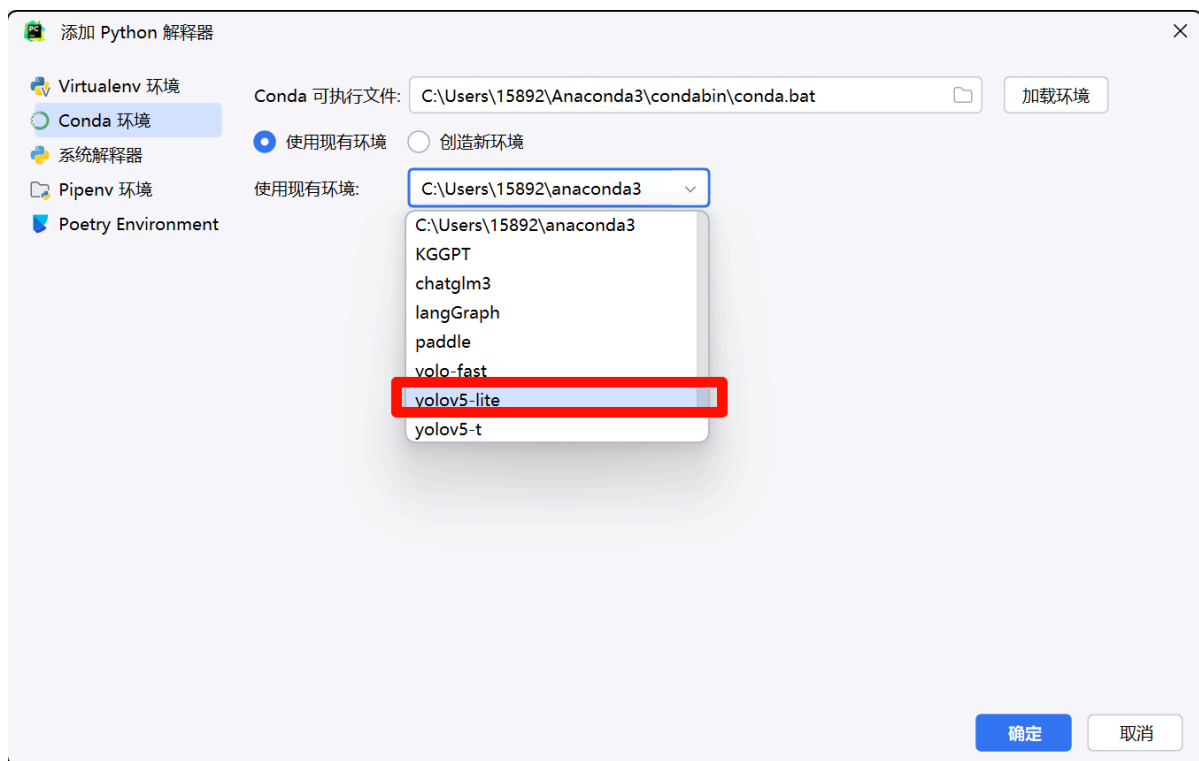
管理软件包...

UTF-8



中





切换完环境后,确保准备好数据集后,我们进行下面的步骤:

我们设置如下几个核心配置:

```
--weights v5lite-s.pt
--cfg models/v5Lite-s.yaml
--data data/coco128.yaml
--img-size 320
--batch-size 16
device 0/cpu
```

Yolov5-Lite 网络模型的训练可以不一定必须使用 **CUDA** 进行加速,但是 **pytorch** 架构等依赖库一定需要满足

讲一下,要修改的地方

data/coco128.yaml:

```
# COCO 2017 dataset http://cocodataset.org - first 128 training images
# Train command: python train.py --data coco128.yaml
# Default dataset location is next to /yolov5:
#   /parent_folder
#   /coco128
#   /yolov5
```

```
# train and val data as 1) directory: path/images/, 2) file: path/images.txt, or
3) list: [path1/images/, path2/images/]
train: 你的训练集地址 # 128 images
val: 你的测试集地址 # 128 images

# number of classes 你的目标有多少类
nc: 4

# class names 目标标签
names: [ '0', '1', '2', '3' ]
```

```
# train and val data as 1) directory: path/images/, 2) file: path/images.txt, or 3) list: [path1/images/, path2/images/]
train: C:/Users/lxd/Desktop/labelme_pic/A/train/images # 你的训练集地址
val: C:/Users/lxd/Desktop/labelme_pic/A/valid/images # 5000 images
#test: C:/Users/lxd/Desktop/labelme_pic/A/test/images/test2017 # 20288 of 40670 images, submit to https://competitions.codalab.org/competiti
# number of classes
nc: 1 需要检测多少类目标
# class names 目标的标签
names: [ 'A' ]
```

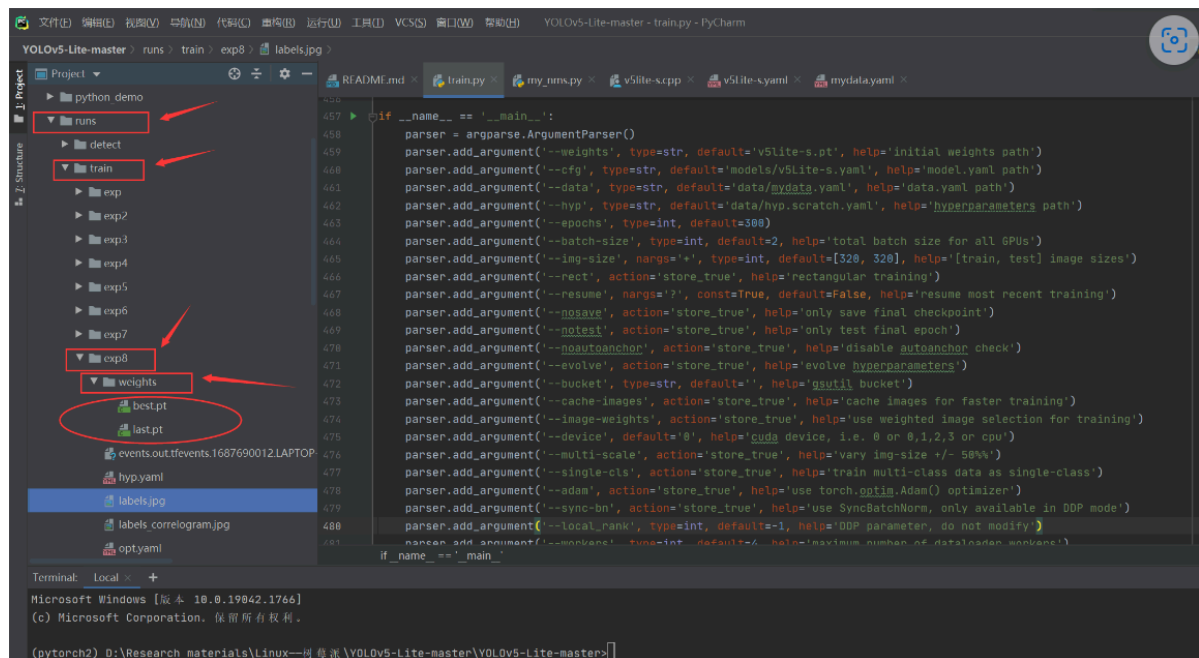
最后在控制塔输入:

```
python train.py --data coco128.yaml --cfg v5lite-s.yaml --weights v5lite-s.pt --
batch-size 16 --device cpu
```

其中device选项调节的是选择cpu训练还是GPU

将训练环境与数据集都搞定之后, 就可以点击运行按钮进行 Yolov5-Lite 的模型训练了!

训练成功之后, 将会在当前目录下的 run 文件下的 trian 文件下找到 expx (x代表数字), expx 则存放了第 x 次训练时候的各种数据内容, 包括: 历史最优权重best_weight, 当前权重last_weight, 训练结果result等等;



模型的转换

我们训练好模型后,要将其转换为onnx格式,在控制台输入如下指令

```
python export.py --weights 你的训练完成后的.pt文件的路径 --end2end
```

即可导出模型!

常见问题

1.

```
ModuleNotFoundError: No module named 'numpy._core'
```

解决方法:升级numpy库 控制台输入:

```
pip install numpy<2
```

2.

```
ModuleNotFoundError: No module named 'requests'
```

解决方法:

```
pip install requests
```

3.

```
AttributeError: module 'numpy' has no attribute 'int'.
`np.int` was a deprecated alias for the builtin `int`. To avoid this error in
existing code, use `int` by itself. Doing this will not modify any behavior and
is safe. When replacing `np.int`, you may wish to use e.g. `np.int64` or
`np.int32` to specify the precision. If you wish to review your current use,
check the release note link for additional information.
The aliases was originally deprecated in NumPy 1.20; for more details and
guidance see the original release note at:
    https://numpy.org/devdocs/release/1.20.0-notes.html#deprecations. Did you
mean: 'inf'?
```

解决方法:

将报错的文件中**np.int**全部修改为 **int**

4.

```
subprocess.CalledProcessError: Command 'git tag' returned non-zero exit status
128.
```

解决方法:

未下载 v5lite-s.pt 模型文件到模型根目录. 找到案例资料中的模型文件移动到目录中即可.

5.

```
RuntimeError: Could not infer dtype of numpy.float32
```

解决方法:

定位报错的文件和语句,显式指定**dtype**: 在将NumPy数组转换为Torch张量时,可以显式指定数据类型。
将其修改为:

```
c = torch.tensor(labels[:, 0], dtype=torch.float32)
```

6.

```
RuntimeError: Numpy is not available
```

解决方法: 控制台输入

```
pip install numpy<2
```